

Occurrence of negative coefficients in Taylor remainder polynomials: The case of skew Dyck paths and HD-free Schröder paths

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ABSTRACT

Eu et al. (2002) [7] introduced new Taylor expansions for the generating functions of the Catalan and Motzkin numbers, establishing the first connection between analytic expansions and enumerative combinatorics. A distinctive feature of such expansions is that the Taylor remainder term is expressed as a polynomial in the generating function itself. However, in all previously studied cases, the corresponding Taylor remainder polynomials contain only nonnegative coefficients, consistent with the nonnegativity of the underlying counting statistics. In contrast, this paper presents the first two combinatorial models whose associated remainder polynomials exhibit negative coefficients, namely skew Dyck paths and HD-free Schröder paths. As a further development, we reveal an equidistribution phenomenon between the two lattice path families with respect to the corresponding step-type statistics. Moreover, combinatorial interpretations for the coefficients of the corresponding Taylor remainder polynomials are presented.

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1. Introduction

Taylor expansion is one of the most fundamental tools in mathematical analysis, providing a convenient way to approximate a sufficiently smooth function by a polynomial at a given point, in particular at the origin. Explicitly, if $f(x)$ is $(n+1)$ -times differentiable in a neighborhood of 0, then its Taylor expansion of order n at 0 is

$$f(x) = f(0) + f'(0)x + \cdots + \frac{f^{(n)}(0)}{n!}x^n + R_n(x),$$

where $R_n(x)$ denotes the remainder term. Several classical forms of the remainder include the Lagrange, Cauchy, and integral forms. These representations play a central role in quantifying the accuracy of the Taylor approximation and in determining the domain on which the approximation remains valid.

Taylor expansions and their remainder terms play an essential role in many areas of mathematics. They provide precise error estimates in numerical approximation and form the basis of Newton-Cotes formulas and polynomial interpolation. In

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Table 1
Selected lattice paths and their generating function equations.

Lattice Paths	Sequences	Functional Equations
Dyck paths	Catalan numbers C_n	$C = 1 + xC^2$
Motzkin paths	Motzkin numbers M_n	$M = 1 + xM + x^2M^2$
Schröder paths	Large Schröder numbers S_n	$S = 1 + xS + x^2S^2$

the study of differential equations, Taylor expansions are used to analyze the local behavior of solutions and to construct convergent power-series representations. In analysis, remainder estimates are indispensable in establishing classical inequalities, asymptotic expansions, and analytic continuation. Furthermore, Taylor approximations appear widely in perturbation methods, optimization, stability analysis, and various computational schemes in applied sciences.

In this work, we examine the role of Taylor expansions and their remainder terms in combinatorial structures. In the classical analytic setting, the remainder depends only on higher-order derivatives of the function. In combinatorial generating functions, by contrast, the remainders in Taylor expansions typically involve the generating function itself. This phenomenon occurs naturally when a generating function satisfies a functional equation that encodes the underlying combinatorial construction. Several examples illustrating this behavior in the context of lattice paths are listed in Table 1, where the Catalan generating function is

$$C = C(x) = \sum_{k \geq 0} C_k x^k,$$

and similarly, the Motzkin and Schröder generating functions are denoted by

$$M = M(x) \quad \text{and} \quad S = S(x),$$

respectively. For background on generating functions and lattice path enumeration, the reader is referred to [17,18].

Eu et al. [7] pioneered the study of Taylor expansions and their corresponding Taylor remainder terms for generating functions arising in combinatorial enumeration models, such as the Catalan generating function $C(x)$ and the Motzkin generating function $M(x)$. Interestingly, these Taylor remainder terms can be expressed as polynomials in the generating function itself. Specifically, they showed that for all $n \geq 1$,

$$C(x) = \sum_{k=0}^{n-1} C_k x^k + x^n f_n(C(x))$$

and

$$M(x) = \sum_{k=0}^{n-1} M_k x^k + x^n g_n(M(x)) + x^{n+1} h_n(M(x)),$$

where $f_n(y)$, $g_n(y)$, and $h_n(y)$ are recursively defined polynomials with nonnegative integer coefficients. In other words, the Taylor expansion of the Catalan generating function has a single-term remainder polynomial, whereas that of the Motzkin generating function has a two-term remainder polynomial. These decompositions have proven particularly useful in revealing the combinatorial structure underlying Catalan and Motzkin numbers, for example, in deriving the enumeration of Dyck paths and Motzkin paths with flaws, as well as providing a simple proof of the Chung–Feller theorem (see also [6] for refined versions).

Furthermore, the new Taylor expansions and their associated remainder polynomials have been generalized in two directions:

- By extending the techniques developed by Eu et al. [7], Liu and Li [11] studied the Taylor expansions of the generating functions of Catalan-like numbers, encompassing the Catalan, Motzkin, and Schröder numbers.
- More recently, Si and Qi [15], two authors of the present paper, studied Taylor expansions whose remainder polynomials consist of a single term, with particular emphasis on applications to 2-Schröder paths, which corresponds to the case $k = 2$ of the k -Schröder paths defined by Ito [8], namely lattice paths of semilength n from $(0, 0)$ to $(2n, 0)$ with step set $\{(i, k-i) \mid i = 0, \dots, k\} \cup \{(1, -1)\}$.

These developments provide a new perspective on analytic remainders and reveal a natural connection between lattice-path combinatorics and the analytic framework of generating functions.

However, to the best of our knowledge, all existing studies on Taylor remainder polynomials involve combinatorial sequences whose coefficients are entirely nonnegative. This naturally raises the following question: since combinatorial models typically give rise to sequences of nonnegative integers, should we also expect the coefficients of their associated Taylor remainder polynomials to be nonnegative?

In this paper, we address this question by studying two combinatorial families. The first is the class of skew Dyck paths, introduced by Deutsch et al. [4], and the second is a newly defined family that we call HD-free Schröder paths. To the best of our knowledge, these constructions provide the first known examples in which the associated Taylor remainder polynomials contain negative coefficients.

Beyond this, we also investigate the combinatorial properties of these two types of lattice paths as well as their interrelations:

- We construct explicit bijections between HD-free Schröder paths and skew Dyck paths, uncovering relationships between statistics of their step types, and thereby establishing their equidistribution.
- We provide combinatorial interpretations of the coefficients of these remainder polynomials, thereby confirming the sign patterns of the coefficients from a combinatorial perspective.

2. A framework for Taylor remainder polynomials

Recently, Si and Qi [15] developed a theoretical framework for Taylor expansions with single-term remainder polynomials, providing a foundation for further exploration of their combinatorial interpretations. This framework extends the study of Taylor remainder polynomials for the Catalan numbers developed by Eu et al. [7].

Lemma 1 ([15]). Let $H(x)$ be a function satisfying $H(x) = 1 + xg(H(x))$, where $g(y) = \sum_{j=0}^m g_j y^j$. Then for any positive integer $n \geq 1$:

(1) The Taylor expansion of $H(x)$ can be written as

$$H(x) = \sum_{k=0}^{n-1} h_k x^k + x^n f_n(H(x)),$$

where $f_n(y) = \sum_{j=0}^{(m-1)n+1} f_{n,j} y^j$.

(2) The coefficients $f_{n+1,j}$ satisfy the recurrence

$$f_{n+1,j} = \sum_{\ell=0}^j \left(\sum_{i=j-\ell+1}^{(m-1)n+1} f_{n,i} \right) g_\ell,$$

with the initial values $f_{1,j} = g_j$ for $0 \leq j \leq m$, and $f_{1,j} = 0$ otherwise.

(3) The coefficients h_k are given by

$$h_k = \sum_{j=0}^{(m-1)k+1} f_{k,j}, \quad k \geq 1, \quad h_0 = 1.$$

Remark 1. The proof by Si and Qi [15] was given under the restriction $g_0 = 0$. However, with a simple adjustment of the argument, the same conclusion remains valid for arbitrary g_0 .

In the next two sections, we introduce two combinatorial families that fit into this framework. The first is the class of skew Dyck paths, and the second is HD-free Schröder paths, a newly defined combinatorial family. As applications of Lemma 1 to these two combinatorial families, we present the corresponding Taylor expansions and remainder polynomials, as well as the recurrence relations for the coefficients of these remainder polynomials.

3. Taylor remainder polynomials for skew Dyck paths

The concept of skew Dyck paths was proposed by Deutsch et al. [4]. Unlike a standard Dyck path, which only allows up steps $U = (1, 1)$ and down steps $D = (1, -1)$, a skew Dyck path additionally permits left-down steps $L = (-1, -1)$. This added flexibility leads to richer combinatorial structures and more diverse enumerative properties, making skew Dyck paths a natural generalization of Dyck paths.

Definition 3.1 (Skew Dyck path [4]). An n -skew Dyck path is a lattice path from $(0, 0)$ to a point on the x -axis consisting of $2n$ steps, where each step is one of $U = (1, 1)$, $D = (1, -1)$, $L = (-1, -1)$, subject to the following conditions:

- The path never goes below the x -axis.
- The patterns UL and LU are both forbidden so as to avoid overlapping steps.

Let $\mathcal{SD}_n$ denote the set of all n -skew Dyck paths, where $n \geq 1$. For example, when $n = 2$, $\mathcal{SD}_2 = \{UUDD, UUDL, UDUD\}$, all of which are illustrated in Fig. 1.

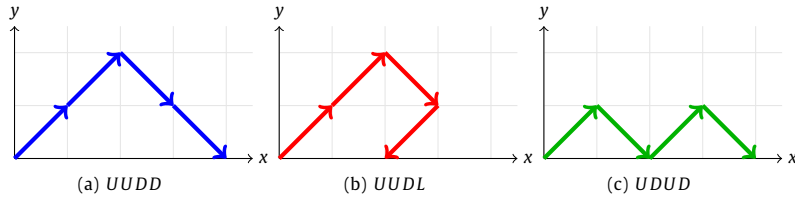


Fig. 1. All 2-skew Dyck paths.

Skew Dyck paths are known to be in bijection with several other combinatorial structures, including hex trees, tree-like polyhexes, 3-Motzkin paths, and various restricted Schröder path families [2,4,5,9,10,13] (see also OEIS sequence [16, A002212]). These connections highlight the rich structural role of skew Dyck paths in enumerative combinatorics. As an illustration, Fig. 2 depicts the correspondence between a tree-like polyhex, its associated hex tree, and the corresponding skew Dyck path. For further enumerative results on skew Dyck paths, the reader is referred to [2].

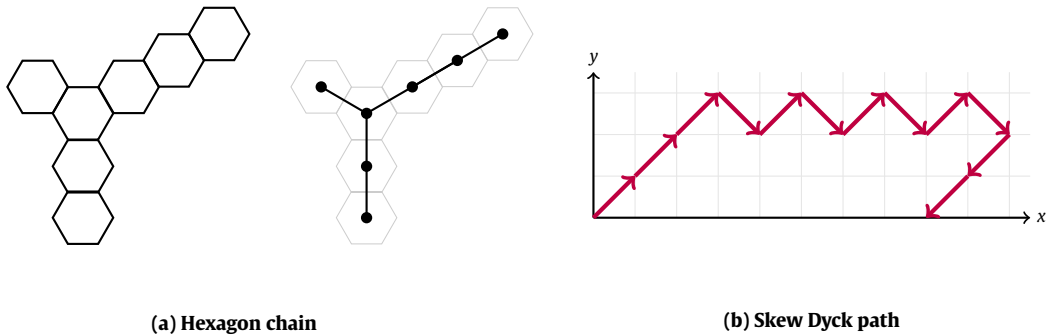


Fig. 2. A tree-like polyhex, corresponding hex tree, and the associated skew Dyck path.

To develop a more general framework for studying their enumerative properties, generating functions, and the associated Taylor remainder polynomials, we associate with each skew Dyck path P three counting statistics: $u(P)$, $d(P)$, and $\ell(P)$, denoting respectively the numbers of up steps U , down steps D , and left-down steps L occurring in P .

For the class of skew Dyck paths, we define the four-variable generating function

$$\mathbf{S}(x, y, z, w) := 1 + \sum_{n \geq 1} w^n \sum_{P \in \mathcal{SD}_n} x^{u(P)} y^{d(P)} z^{\ell(P)}.$$

For each $n \geq 1$, we define

$$\mathbf{s}_n(x, y, z) := \sum_{P \in \mathcal{SD}_n} x^{u(P)} y^{d(P)} z^{\ell(P)},$$

and set $\mathbf{s}_0(x, y, z) = 1$, so that $\mathbf{s}_n(x, y, z)$ is exactly the coefficient of w^n in the expansion of $\mathbf{S}(x, y, z, w)$.

Thus, the variables x , y , and z may be regarded as the weights assigned to the steps U , D , and L of a skew Dyck path, respectively. For brevity, we often write $\mathbf{s}_n$ and $\mathbf{S}(w)$ when the variables (weights) x, y, z are omitted without causing ambiguity. In particular,

$$\mathbf{S}(w) = \sum_{n \geq 0} \mathbf{s}_n w^n.$$

The first few values of s_n are

$$s_0 = 1,$$

$$s_1 = xy,$$

$$s_2 = 2x^2y^2 + x^2yz,$$

$$s_3 = 5x^3y^3 + 4x^3y^2z + x^3yz^2,$$

$$s_4 = 14x^4y^4 + 15x^4y^3z + 6x^4y^2z^2 + x^4yz^3,$$

$$s_5 = 42x^5y^5 + 56x^5y^4z + 30x^5y^3z^2 + 8x^5y^2z^3 + x^5yz^4,$$

$$s_6 = 132x^6y^6 + 210x^6y^5z + 140x^6y^4z^2 + 50x^6y^3z^3 + 10x^6y^2z^4 + x^6yz^5.$$

A standard decomposition of any nonempty skew Dyck path leads to the following functional equation for the generating function $S(w)$ (for an introduction to the kernel method, see [14]), along with its explicit expression.

Theorem 2. Let $S(w)$ be the generating function of skew Dyck paths. Then $S(w)$ satisfies the functional equation

$$S(w) = 1 + xyw S(w)^2 + xzw (S(w) - 1). \quad (1)$$

Moreover, $S(w)$ can be expressed explicitly as

$$S(w) = \frac{1 - xzw - \sqrt{(1 - xzw)(1 - xzw - 4xyw)}}{2xyw}. \quad (2)$$

Proof. The functional equation (1) follows from the standard decomposition of skew Dyck paths as follows:

- the empty path (i.e., $n = 0$), which contributes the term $s_0 w^0 = 1$;
- a path whose first return to the x -axis is achieved by a down step D (see Fig. 3, left panel);
- otherwise, the first return to the x -axis occurs via a left-down step L , which must be the last step of the path. In this case, the initial up step U cannot be immediately followed by this final L , hence the path contains at least one step between its first and last steps (see Fig. 3, right panel).

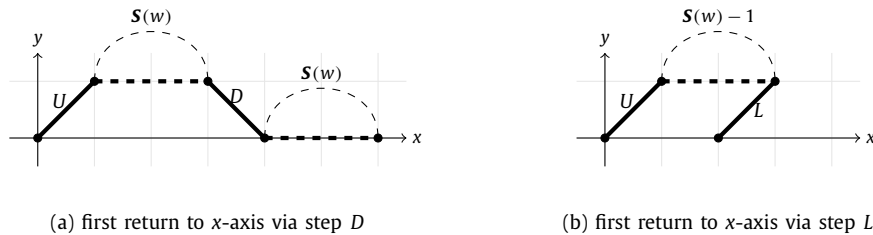


Fig. 3. Decomposition of skew Dyck paths.

This decomposition immediately leads to the functional equation (1). The explicit expression (2) is then obtained by solving the quadratic equation (1). $\square$

Equation (1) fits into the framework of Lemma 1. Indeed,

$$S(w) = 1 + w(xy S(w)^2 + xz S(w) - xz) = 1 + w g(S(w)),$$

where $g(u) = xy u^2 + xz u - xz$. Let g_0 denote the term of $g(u)$ that is independent of u . In this case, $g_0 = -xz$, which is nonzero. This feature distinguishes our setting from previous studies on Taylor remainder polynomials for other combinatorial objects, such as Dyck paths, Schröder paths, and 2-Schröder paths [7,11,15], where the corresponding constant term always vanishes, i.e., $g_0 = 0$.

Applying Lemma 1 to $S(w)$ with $g(u) = xy u^2 + xz u - xz$, we immediately conclude that $S(w)$ admits a Taylor expansion with a single-term remainder polynomial. Moreover, this substitution yields explicit recurrence relations for the coefficients of the remainder polynomial.

Theorem 3. Let $S(w)$ be the generating function defined by the skew Dyck paths. For any integer $n \geq 1$,

(1) $\mathbf{S}(w)$ admits a Taylor expansion

$$\mathbf{S}(w) = \sum_{k=0}^{n-1} \mathbf{s}_k w^k + w^n F_n(\mathbf{S}(w)),$$

where $F_n(u)$ is the n -th Taylor remainder polynomial.

(2) Let $f_{n,j}(x, y, z)$, which we simply write as $f_{n,j}$ for brevity, denote the coefficient of u^j in $F_n(u)$, so that $F_n(u) = \sum_{j=0}^{n+1} f_{n,j} u^j$. Then the family $\{f_{n,j}\}$ satisfies the following recurrence relations:

$$\begin{aligned} f_{n+1,0} &= -xz \sum_{i=1}^{n+1} f_{n,i}, & f_{n+1,1} &= (xz)^{n+1}, \\ f_{n+1,j} &= xz f_{n,j} + xy \sum_{i=j-1}^{n+1} f_{n,i}, & \text{for } 2 \leq j \leq n+2, \end{aligned}$$

with initial conditions $f_{1,0} = -xz$, $f_{1,1} = xz$, $f_{1,2} = xy$, and $f_{1,j} = 0$ otherwise.

(3) The coefficient $\mathbf{s}_n$ of w^n in $\mathbf{S}(w)$ is obtained as

$$\mathbf{s}_n = \sum_{j=0}^{n+1} f_{n,j}, \quad n \geq 1, \quad \mathbf{s}_0 = 1.$$

Remark 2. From the recurrence relations and initial conditions for $f_{n,j}$, it follows that all coefficients of $f_{n,j}$ in the variables x, y, z are nonnegative for $1 \leq j \leq n+1$, whereas all coefficients of $f_{n,0}$ are nonpositive. To the best of our knowledge, this provides the first combinatorial model whose Taylor remainder polynomial contains a negative coefficient. In contrast, the Taylor remainder coefficients for Dyck paths, Motzkin paths, Schröder paths, and 2-Schröder paths are all nonnegative [7,11,15].

For example, we list the n -th Taylor remainder polynomials $F_n(u)$ for $n = 1, 2, 3$:

$$\begin{aligned} F_1(u) &= -xz + xz u + xy u^2, \\ F_2(u) &= -x^2 yz - x^2 z^2 + x^2 z^2 u + (x^2 y^2 + 2x^2 yz) u^2 + x^2 y^2 u^3, \\ F_3(u) &= -2x^3 y^2 z - 2x^3 yz^2 - x^3 z^3 + x^3 z^3 u + (2x^3 y^3 + 3x^3 y^2 z + 3x^3 yz^2) u^2 \\ &\quad + (2x^3 y^3 + 3x^3 y^2 z) u^3 + x^3 y^3 u^4. \end{aligned}$$

Under the setting $z = -y$, additional negative coefficients appear. The ratios of the resulting $f_{n,j}$ to $(xy)^n$ for $1 \leq n \leq 10$ can then be arranged into the following triangular array:

1	-1	1							
0	1	-1	1						
1	-1	2	-1	1					
1	1	-1	3	-1	1				
3	-1	4	-1	4	-1	1			
6	1	2	8	-1	5	-1	1		
15	-1	13	6	13	-1	6	-1	1	
36	1	23	31	11	19	-1	7	-1	1
91	-1	68	59	56	17	26	-1	8	-1
232	1	164	174	109	89	24	34	-1	9

The numerical values of these coefficients exhibit several interesting patterns:

- The first column coincides with the Riordan numbers [3], whose generating function is $\frac{1+x-\sqrt{1-2x-3x^2}}{2(1+x)}$.
- The parity pattern in the third column matches the signs of the Thue–Morse sequence [1], whose generating function is $\frac{1+x-\sqrt{1-2x-3x^2}}{2(1+x)^2}$.
- All negative entries are equal to -1 . Moreover, except for the -1 's appearing in the second column, the remaining -1 's occur exclusively along two diagonal lines.

These observations suggest that it would be interesting to further investigate and characterize the combinatorial structure underlying this triangular array.

Next, we naturally wish to understand whether the coefficients $f_{n,j}$ of the Taylor remainder polynomials admit combinatorial interpretations via an appropriate model. In the setting of skew Dyck paths, the presence of left-down steps L makes the path's behavior difficult to control. Consequently, when seeking combinatorial interpretations of $f_{n,j}$, we need to employ alternative paths that are easier to handle. To this end, we introduce a new family of lattice paths, which we call HD-free Schröder paths. We then construct an explicit bijection between these paths and skew Dyck paths, and characterize the relationship between their respective step-type statistics.

4. Taylor remainder polynomials for HD-free Schröder paths

We introduce a new restriction on classical Schröder paths which, to the best of our knowledge, has not appeared previously in the literature.

Definition 4.1 (*HD-free Schröder path*). An HD-free Schröder path of semilength n is a lattice path from $(0, 0)$ to $(2n, 0)$ using steps $U = (1, 1)$, $D = (1, -1)$, and $H = (2, 0)$, never going below the x -axis, and satisfying the following conditions:

- a horizontal step H is not immediately followed by a down step D (the ‘HD’ restriction);
- the path does not end with an H step.

The HD restriction distinguishes this family from classical Schröder paths [8,15,19]. It forces each H step to be followed by either a U step or another H step, thereby producing horizontal segments that are never immediately followed by a descent. This structural constraint is precisely what allows us to construct a bijection with the skew Dyck paths discussed earlier. Moreover, we can refine their enumerative equivalence by taking step-type statistics into account.

Let $\mathcal{HDFS}_n$ denote the set of all HD-free Schröder paths of semilength n , where $n \geq 1$. For example, when $n = 2$, $\mathcal{HDFS}_2 = \{UUDD, HUD, UDUD\}$, as illustrated in Fig. 4.

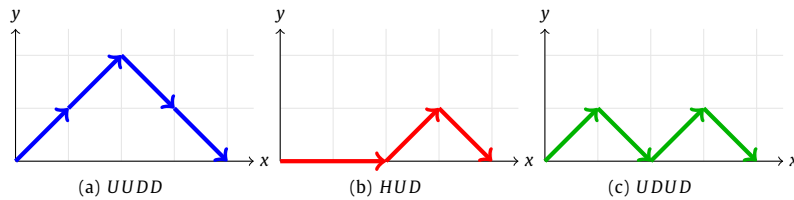


Fig. 4. All HD-free Schröder paths of semilength 2.

Comparing Figs. 1 and 4, we observe that the number of 2-skew Dyck paths coincides with the number of HD-free Schröder paths of semilength 2. This observation naturally suggests an enumerative equivalence between HD-free Schröder paths of semilength n and n -skew Dyck paths, which will be confirmed later.

In the setting of HD-free Schröder paths, each path P carries three natural step statistics: $u(P)$, $d(P)$, and $h(P)$, which record the numbers of up steps U , down steps D , and horizontal steps H , respectively. To encode these statistics simultaneously, we introduce the multivariate generating function

$$\mathbf{H}(x, y, z, w) := 1 + \sum_{n \geq 1} w^n \sum_{P \in \mathcal{HDFS}_n} x^{u(P)} y^{d(P)} z^{h(P)}.$$

For convenience, we denote the inner sum by

$$\mathbf{h}_n(x, y, z) := \sum_{P \in \mathcal{HDFS}_n} x^{u(P)} y^{d(P)} z^{h(P)}$$

for $n \geq 1$, and set $\mathbf{h}_0(x, y, z) = 1$, so that $\mathbf{h}_n(x, y, z)$ is the coefficient of w^n in the series expansion of $\mathbf{H}(x, y, z, w)$.

In situations where the step weights x , y , and z play no role, we suppress them from the notation and simply write $\mathbf{h}_n$ and $\mathbf{H}(w)$. Under this convention, the generating function reduces to

$$\mathbf{H}(w) = \sum_{n \geq 0} \mathbf{h}_n w^n.$$

A standard decomposition of nonempty HD-free Schröder paths gives rise to the following functional equation for the multivariate generating function $\mathbf{H}(w)$ and its explicit closed form.

Theorem 4. Let $H(w)$ be the generating function of HD-free Schröder paths. Then $H(w)$ satisfies the functional equation

$$H(w) = 1 + xyw H(w)^2 + zw (H(w) - 1). \quad (3)$$

Furthermore, $H(w)$ admits the explicit closed-form expression

$$H(w) = \frac{1 - zw - \sqrt{(1 - zw)(1 - zw - 4xyw)}}{2xyw}. \quad (4)$$

Proof. We apply a standard first-step decomposition. Let P be a nonempty HD-free Schröder path. Its first step is either U or H .

Case 1: The first step is U .

In this case, the path admits the unique decomposition $P = U P_1 D P_2$, where P_1 and P_2 are (possibly empty) HD-free Schröder paths, see Fig. 5(a). The contribution of the two steps U , D , and the length increase is xyw , and the two subpaths contribute multiplicatively. Thus this case contributes $xyw H(w)^2$.

Case 2: The first step is H .

An HD-free Schröder path can not end with an H step. Hence, after an initial H , the remainder must be a nonempty HD-free Schröder path, see Fig. 5(b). Since the H step has weight z and contributes one unit of semilength, the total contribution of this case is $zw(H(w) - 1)$.

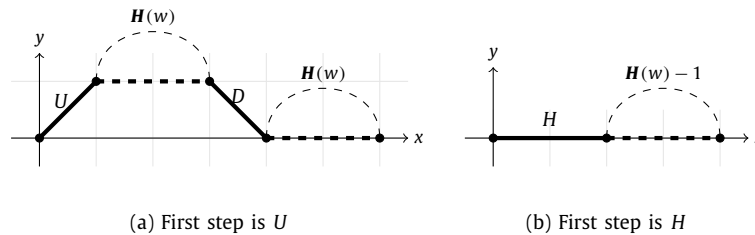


Fig. 5. Decomposition of HD-free Schröder paths based on the initial step type.

Including the empty path, which contributes 1, we obtain the functional equation (3). Solving this quadratic equation for $H(w)$ then leads to the closed-form expression given in Equation (4). $\square$

Remark 3. A direct inspection of Equation (2) and Equation (4) shows that the generating functions $S(x, y, z, w)$ and $H(x, y, z, w)$ agree under the substitution

$$S(x, y, z, w) = H(x, y, xz, w). \quad (5)$$

Consequently, the corresponding coefficient polynomials coincide under the same substitution:

$$s_n(x, y, z) = h_n(x, y, xz), \quad \text{for all } n \geq 0.$$

In particular, by setting $x = y = z = 1$, the above identity reduces to a counting correspondence between the number of skew Dyck paths with $2n$ steps and the number of HD-free Schröder paths of semilength n , for every $n \geq 1$.

Theorem 5. For every $n \geq 1$, the number of skew Dyck paths with $2n$ steps coincides with the number of HD-free Schröder paths of semilength n .

Furthermore, considering the functional equation (3) satisfied by $H(w)$, the generating function can be seen as conforming to the framework of Lemma 1. Applying Lemma 1 to $H(w)$ with $g(u) = xyu^2 + zu - z$ shows that its Taylor expansion has a remainder consisting of a single term. Moreover, this substitution yields explicit recurrence relations for the coefficients of the remainder polynomial.

Theorem 6. Let $H(w)$ be the generating function of HD-free Schröder paths. For any integer $n \geq 1$,

(1) $H(w)$ admits a Taylor expansion

$$H(w) = \sum_{k=0}^{n-1} h_k w^k + w^n R_n(H(w)),$$

where $R_n(u)$ is the n -th Taylor remainder polynomial.

Table 2Examples illustrating the equidistribution for $n = 3$.

(i, j, k)	$\mathcal{SD}_3$	$\mathcal{HDFS}_3$
$(3, 3, 0)$	$\{UUUDDD, UUDDUD, UDUUDD, UDUDUD, UUDUDD\}$	$\{UUUDDD, UUDDUD, UDUUDD, UDUDUD, UUDUDD\}$
$(3, 2, 1)$	$\{UUUDL, UDUUDL, UUUDDL, UUUDDL\}$	$\{HUUUD, HDUUD, HUDD, UHUU\}$
$(3, 1, 2)$	$\{UUUDDL\}$	$\{HHUD\}$

- (2) Let $r_{n,j}(x, y, z)$, written as $r_{n,j}$ for short, denote the coefficient of u^j in $R_n(u)$, namely, $R_n(u) = \sum_{j=0}^{n+1} r_{n,j} u^j$. Then $r_{n,j}$ satisfies the following recurrence relations:

$$r_{n+1,0} = -z \sum_{i=1}^{n+1} r_{n,i}, \quad r_{n+1,1} = z^{n+1},$$

$$r_{n+1,j} = z r_{n,j} + xy \sum_{i=j-1}^{n+1} r_{n,i}, \quad \text{for } 2 \leq j \leq n+2,$$

with initial conditions $r_{1,0} = -z$, $r_{1,1} = z$, $r_{1,2} = xy$, and $r_{1,j} = 0$ otherwise.

- (3) The coefficient $\mathbf{h}_n$ of w^n in $\mathbf{H}(w)$ is recovered by

$$\mathbf{h}_n = \sum_{j=0}^{n+1} r_{n,j}, \quad n \geq 1, \quad \mathbf{h}_0 = 1.$$

By Equation (5), we have $\mathbf{S}(x, y, z, w) = \mathbf{H}(x, y, xz, w)$, or equivalently, $\mathbf{S}(w) = \mathbf{H}(w)$. Consequently, the Taylor remainder polynomial $F_n(u)$ associated with $\mathbf{S}(w)$ transforms into the Taylor remainder polynomial $R_n(u)$ associated with $\mathbf{H}(w)$ by the substitution $xz \mapsto z$.

Based on the illustrative cases $n = 1, 2, 3$ of the Taylor remainder polynomials $F_n(u)$ given earlier, we derive the following Taylor remainder polynomials $R_n(u)$:

$$R_1(u) = -z + zu + xy u^2,$$

$$R_2(u) = -xyz - z^2 + z^2 u + (x^2 y^2 + 2xyz) u^2 + x^2 y^2 u^3,$$

$$R_3(u) = -2x^2 y^2 z - 2xyz^2 - z^3 + z^3 u + (2x^3 y^3 + 3x^2 y^2 z + 3xyz^2) u^2$$

$$+ (2x^3 y^3 + 3x^2 y^2 z) u^3 + x^3 y^3 u^4.$$

5. Bijection between skew Dyck paths and HD-free Schröder paths

From an algebraic perspective, we have established the counting equivalence between skew Dyck paths and HD-free Schröder paths via their generating functions. More specifically, the relation between their generating functions given in (5) suggests an equidistribution of the corresponding step-type statistics across the two combinatorial families.

Theorem 7. Define the statistics

$$\alpha(P) = (u(P) + h(P), d(P), h(P)), \quad P \in \mathcal{HDFS}_n,$$

$$\beta(P') = (u(P'), d(P'), \ell(P')), \quad P' \in \mathcal{SD}_n.$$

Then α on $\mathcal{HDFS}_n$ and β on $\mathcal{SD}_n$ are equidistributed, i.e.,

$$\#\{P \in \mathcal{HDFS}_n : \alpha(P) = (i, j, k)\} = \#\{P' \in \mathcal{SD}_n : \beta(P') = (i, j, k)\}$$

for any triple (i, j, k) .

As an example, we consider the case $n = 3$ in Table 2, where we list all triples (i, j, k) for which both $\mathcal{HDFS}_3$ and $\mathcal{SD}_3$ are nonempty.

We now proceed to construct an explicit bijection between these two combinatorial families by suitably transforming their step types, thereby providing a combinatorial proof of the aforementioned equidistribution property.

The main difference between standard Dyck paths and Schröder paths is that Schröder paths allow horizontal steps while Dyck paths do not. Such a fundamental distinction also appears in their restricted versions, namely skew Dyck paths and HD-free Schröder paths. Moreover, skew Dyck paths permit the use of left-down steps L , whereas the HD-free condition forbids a horizontal step H in an HD-free Schröder path from being immediately followed by a down step D .

Thus, in order to construct a bijection between skew Dyck paths and HD-free Schröder paths, it is crucial to determine how to handle the horizontal steps in an HD-free Schröder path. In essence, we establish a one-to-one correspondence by replacing each horizontal step H (of length 2) in an HD-free Schröder path with a pair of steps, an up step U and a left-down step L , in the corresponding skew Dyck path.

Lemma 8. For any $n \geq 1$, there exists a bijection $\phi : \mathcal{HDFS}_n \rightarrow \mathcal{SD}_n$. In particular, if $P \in \mathcal{HDFS}_n$ and $P' = \phi(P)$, then

$$u(P') = u(P) + h(P), \quad d(P') = d(P), \quad \ell(P') = h(P).$$

By Lemma 8, the bijection $\phi : \mathcal{HDFS}_n \rightarrow \mathcal{SD}_n$ sends each path $P \in \mathcal{HDFS}_n$ to a path $P' = \phi(P) \in \mathcal{SD}_n$ such that $u(P') = u(P) + h(P)$, $d(P') = d(P)$, and $\ell(P') = h(P)$. Consequently,

$$x^{u(P')} y^{d(P')} z^{\ell(P')} = x^{u(P)} y^{d(P)} (xz)^{h(P)}.$$

Summing over all paths of size n and then over all $n \geq 0$, we obtain

$$\sum_{n \geq 0} w^n \sum_{P' \in \mathcal{SD}_n} x^{u(P')} y^{d(P')} z^{\ell(P')} = \sum_{n \geq 0} w^n \sum_{P \in \mathcal{HDFS}_n} x^{u(P)} y^{d(P)} (xz)^{h(P)},$$

which results in $\mathbf{S}(x, y, z, w) = \mathbf{H}(x, y, xz, w)$ again.

The following bijection is defined uniformly for paths of all lengths,

$$\phi : \mathcal{HDFS}_n \rightarrow \mathcal{SD}_n, \quad \phi^{-1} : \mathcal{SD}_n \rightarrow \mathcal{HDFS}_n, \quad \text{for every } n \geq 1,$$

thereby completing the proof of Lemma 8. A key feature of the bijection between $\mathcal{HDFS}_n$ and $\mathcal{SD}_n$ is that each horizontal step H in an HD-free Schröder path corresponds uniquely to a pair of steps U and L in the associated skew Dyck path, and vice versa. This one-to-one correspondence enables us to translate step statistics directly between the two families.

When $n = 1$, the statement is immediate, since $\mathcal{HDFS}_1 = \mathcal{SD}_1 = \{UD\}$. For $n = 2$, Fig. 1 illustrates the three paths in $\mathcal{SD}_n$, while Fig. 4 depicts the three paths in $\mathcal{HDFS}_n$. We define the bijection ϕ by mapping the i -th path in Fig. 4 to the i -th path in Fig. 1, for $i = 1, 2, 3$. In what follows, we assume that $n \geq 3$.

Assume that $P \in \mathcal{HDFS}_n$. If $h(P) = 0$, then P contains no horizontal steps H and is itself a (skew) Dyck path. In this case, we define the bijection simply by $\phi(P) = P$.

Now assume that $h(P) \geq 1$. Let the leftmost horizontal step of P , denoted by H^* , occur at height k . Then P admits a unique decomposition $P = Q_1 H^* Q_2 Q_3$, where:

- Q_1 is the longest initial segment of P containing no horizontal steps. This segment may be empty, for example, if P begins with a horizontal step.
- H^* is the leftmost horizontal step of P .
- Q_2 is the longest segment immediately following H^* whose steps never drop below height k . By the HD-free condition, this segment cannot be empty.
- Q_3 is the remaining suffix of P . This segment may be empty, for instance, when H^* lies on the x -axis (i.e., at height 0).

Let us define $\phi_1(P) = Q_1 U^* Q_2^* L^* Q_3^*$, where:

- The segment Q_1 in P and in $\phi_1(P)$ is identical.
- U^* and L^* are new steps corresponding to the leftmost horizontal step H^* of P .
- Q_2^* is the path obtained from Q_2 of P by shifting it upward by 1 unit and leftward by 1 unit.
- Q_3^* is the path obtained from Q_3 of P by shifting it leftward by 1 unit. Note that Q_3^* is empty if Q_3 does not exist.

The correspondence between P and $\phi_1(P)$ is illustrated in Figs. 6 and 7, where the former corresponds to $k = 0$ and the latter to $k > 0$.

It is clear that

$$\begin{aligned} u(\phi_1(P)) &= u(P) + 1, & d(\phi_1(P)) &= d(P), \\ h(\phi_1(P)) &= h(P) - 1, & \ell(\phi_1(P)) &= 1. \end{aligned}$$

Similarly, we can define the maps $\phi_2, \phi_3, \dots, \phi_{h(P)}$, each time splitting the leftmost horizontal step into a new pair of up step U and left-down step L , until no horizontal steps remain. At this point, the path becomes a skew Dyck path.

Finally, we define $\phi = \phi_{h(P)} \circ \phi_{h(P)-1} \circ \dots \circ \phi_1$, so that $\phi(P) \in \mathcal{SD}_n$. This gives the desired map from $\mathcal{HDFS}_n$ to $\mathcal{SD}_n$. Moreover,

$$u(\phi(P)) = u(P) + h(P), \quad d(\phi(P)) = d(P), \quad \ell(\phi(P)) = h(P).$$

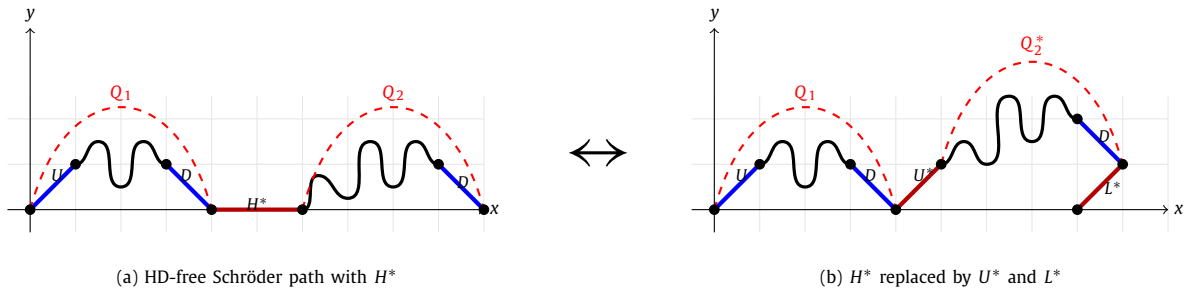


Fig. 6. The horizontal step H^* lies on the x -axis.



Fig. 7. The horizontal step H^* lies strictly above the x -axis.

The inverse map ϕ^{-1} can be defined in a completely analogous way. Since ϕ^{-1} maps skew Dyck paths to HD-free Schröder paths, for any $P' \in \mathcal{SD}_n$ we set $P = \phi^{-1}(P') \in \mathcal{HDFS}_n$. The construction proceeds by systematically eliminating all left-down steps L in the skew Dyck path P' , thereby producing the HD-free Schröder path P .

If $\ell(P') = 0$, then P' is both a skew Dyck path and an HD-free Schröder path. In this case we simply set $P = P'$, that is, $P = \phi^{-1}(P') = P'$.

Otherwise $\ell(P') > 0$. In this case P' admits a unique decomposition $P' = R_1 U^* R_2 L^* R_3$, where

- L^* is the rightmost left-down step of P' ;
- U^* is the up step located to the left of L^* at the same height, and it is the closest such up step;
- R_2 is the segment between U^* and L^* , hence every step in R_2 lies strictly above the common height of U^* and L^* . Note that R_2 is nonempty, since by the definition of a skew Dyck path, an up step U cannot be immediately followed by a left-down step L ;
- R_1 is the initial segment preceding U^* , it may be empty (for instance, when U^* is the first step);
- R_3 is the final segment following L^* , it may also be empty (for instance, when L^* is the last step).

We define the mapping $\varphi_1(P') = R_1 H^* R_2^* R_3^*$, where:

- H^* is a new horizontal step corresponding to the pair (U^*, L^*) in the path P' ;
- R_1 is identical to the segment R_1 in the decomposition of P' ;
- R_2^* is obtained from R_2 of P' by shifting every step downward by 1 unit and to the right by 1 unit;
- R_3^* is obtained from R_3 of P' by shifting every step to the right by 2 units.

The correspondence between P' and $\varphi_1(P')$ is illustrated in Figs. 6 and 7, depending on whether U^* and L^* lie on the x -axis or strictly above it. In each figure, P' is shown on the right and $\varphi_1(P')$ on the left, obtained by adjusting the notation in the indicated segments.

It is easy to verify that

$$\begin{aligned} u(\varphi_1(P')) &= u(P') - 1, & d(\varphi_1(P')) &= d(P'), \\ \ell(\varphi_1(P')) &= \ell(P') - 1, & h(\varphi_1(P')) &= 1. \end{aligned}$$

Analogously, we define the maps $\varphi_2, \varphi_3, \dots, \varphi_{\ell(P')}$, each of which removes a pair of up step U and left-down step L from the current path. We continue this process until no left-down steps remain, eventually obtaining an HD-free Schröder path.

Table 3HD-free Schröder subpaths and their generating functions for $n = 1$.

j	$S_U(1, j)$	$S_{1,j;U}$	$S_H(1, j)$	$S_{1,j;H}$	$S(1, j)$	$S_{1,j}$
0	$\emptyset$	0	$\{H\}$	z	$\{H\}$	z
1	$\{U\}$	x	$\emptyset$	0	$\{U\}$	x

Formally, set $\phi^{-1} = \varphi_{\ell(P')} \circ \varphi_{\ell(P')-1} \circ \cdots \circ \varphi_1$, so that $P = \phi^{-1}(P')$ is an HD-free Schröder path. Moreover, the statistics satisfy

$$u(P) = u(P') - \ell(P'), \quad d(P) = d(P'), \quad h(P) = \ell(P').$$

Thus, we have constructed a bijection between $\mathcal{HDFS}_n$ and $\mathcal{SD}_n$ for every $n \geq 1$, thereby completing the proof of Lemma 8.

For example, we may refer to the skew Dyck path and its corresponding HD-free Schröder path shown in Fig. 8.

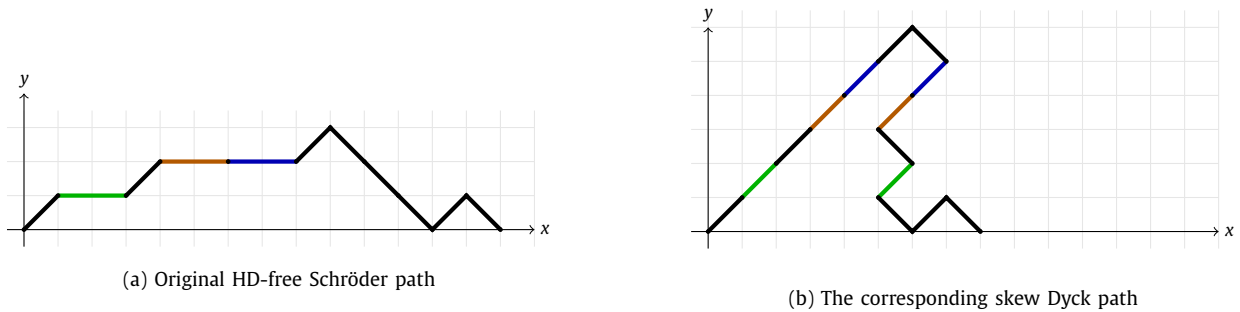


Fig. 8. An example illustrating the correspondence between an HD-free Schröder path and a skew Dyck path, where the colored horizontal step in the left figure corresponds to a pair of an up step and a left-down step in the right figure.

6. Combinatorial interpretations of Taylor remainder polynomial coefficients

We now give a combinatorial interpretation of the coefficients $f_{n,j}$ and $r_{n,j}$ of the Taylor remainder polynomials for the generating functions $S(w)$ and $H(w)$, corresponding to skew Dyck paths and HD-free Schröder paths, respectively. Noted that $f_{n,j}$ and $r_{n,j}$ are polynomials in the variables x, y, z .

6.1. Combinatorial interpretations of $r_{n,j}$

To this end, we first introduce some definitions for HD-free Schröder subpaths starting at $(0, 0)$ and ending at their first intersection with a line of the form $u + v = 2n$, for any integer $n \geq 1$. More precisely, for integers $n \geq 1$ and $0 \leq j \leq n$, define $S(n, j)$ to be the set of HD-free Schröder subpaths from $(0, 0)$ to $(2n - j, j)$ that first meet the line $u + v = 2n$ at their endpoint.

It is easy to observe that, under the “first-intersection” requirement, such HD-free Schröder subpaths must end with either an up step U or a horizontal step H ; in other words, it is impossible for them to end with a down step D . Therefore, we further partition $S(n, j)$ into two subsets, $S_U(n, j)$ and $S_H(n, j)$, consisting of those subpaths whose first intersection with the line $u + v = 2n$ occurs at their endpoint via an up step U and a horizontal step H , respectively. For the boundary values $j \in \{-1, n + 1\}$, we adopt the convention that $S(n, j) = S_U(n, j) = S_H(n, j) = \emptyset$.

Further, for integers $n \geq 1$ and $0 \leq j \leq n$, define

$$S_{n,j}(x, y, z) = \sum_{P \in S(n,j)} x^{u(P)} y^{d(P)} z^{h(P)},$$

$$S_{n,j;U}(x, y, z) = \sum_{P \in S_U(n,j)} x^{u(P)} y^{d(P)} z^{h(P)},$$

$$S_{n,j;H}(x, y, z) = \sum_{P \in S_H(n,j)} x^{u(P)} y^{d(P)} z^{h(P)}.$$

When no confusion can arise, we simply write $S_{n,j}$, $S_{n,j;U}$, and $S_{n,j;H}$. To facilitate understanding, the cases $n = 1$ and $n = 2$ are presented in Tables 3 and 4, respectively. See also Fig. 9 for an illustration of the paths when $n = 2$.

Table 4
HD-free Schröder subpaths and their generating functions for $n = 2$.

j	$S_U(2, j)$	$S_{2,j;U}$	$S_H(2, j)$	$S_{2,j;H}$	$S(2, j)$	$S_{2,j}$
0	$\emptyset$	0	$\{UDH, HH\}$	$xyz + z^2$	$\{UDH, HH\}$	$xyz + z^2$
1	$\{HU, UDU\}$	$xz + x^2y$	$\{UH\}$	xz	$\{HU, UDU, UH\}$	$2xz + x^2y$
2	$\{UU\}$	x^2	$\emptyset$	0	$\{UU\}$	x^2

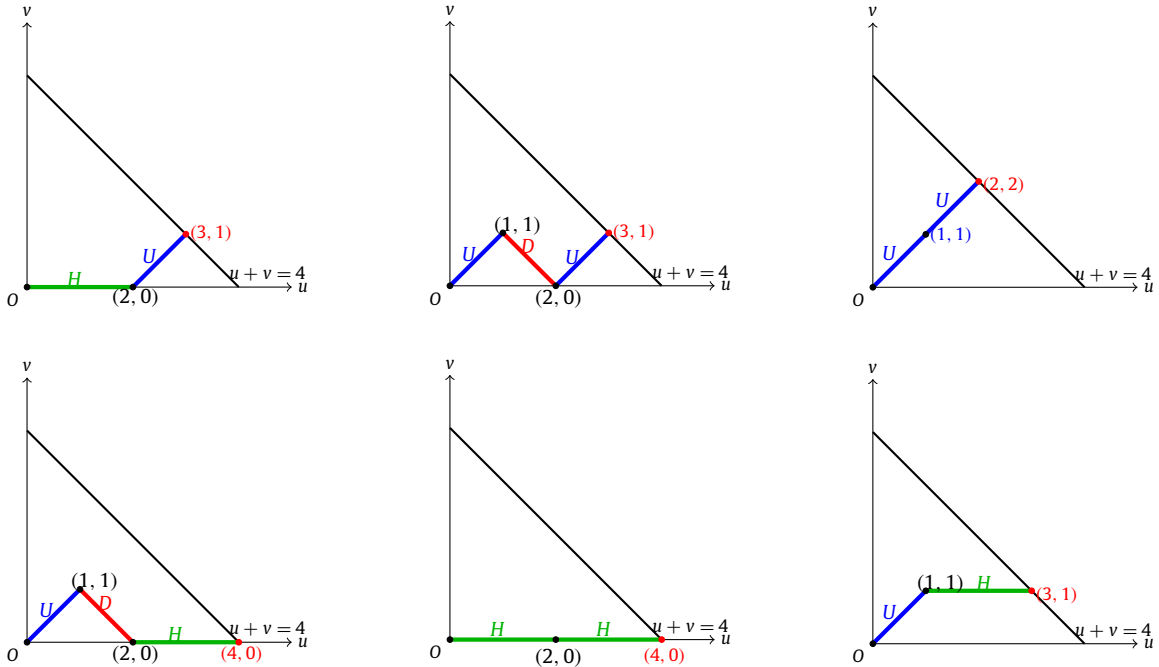


Fig. 9. The paths in $S_U(2, j)$ and $S_H(2, j)$ for $j = 0, 1, 2, 3$.

The initial conditions for $S_{n,j;U}$ and $S_{n,j;H}$ can be read directly from Table 3 for the case $n = 1$:

$$S_{1,j;U} = \begin{cases} x, & \text{if } j = 1, \\ 0, & \text{if } j \neq 1, \end{cases} \quad S_{1,j;H} = \begin{cases} z, & \text{if } j = 0, \\ 0, & \text{if } j \neq 0. \end{cases}$$

The coupled recurrence relations for $S_{n,j;U}$ and $S_{n,j;H}$ follow from the standard decomposition of paths in $S_U(n, j)$ and $S_H(n, j)$, respectively. Each path $P \in S_U(n, j)$ with $j \geq 1$ can be decomposed into three parts $P = P_1 P_2 U$, where P_1 is the initial segment of P up to the first time it meets the line $u + v = 2n - 2$, namely at the point $(2n - 2 - i, i)$ for some i with $j - 1 \leq i \leq n - 1$. The segment P_2 then consists of $i - j + 1$ consecutive D -steps along the line $u + v = 2n - 2$, and the final step U goes from $(2n - j - 1, j - 1)$ to $(2n - j, j)$. We remark that, due to the HD-free restriction, if the final step of P_1 is a horizontal step H , then P_1 necessarily meets the line $u + v = 2n - 2$ at the point $(2n - 1 - j, j - 1)$, and then no such path P_2 exists. Hence,

$$S_{n,j;U} = x \left(S_{n-1,j-1;H} + \sum_{i=j-1}^{n-1} S_{n-1,i;U} y^{i-j+1} \right)$$

for $1 \leq j \leq n$. In particular, $S_{n,0;U} = 0$. Similarly, by considering paths in $S_H(n, j)$ whose last step is H , we obtain

$$S_{n,j;H} = z \left(S_{n-1,j;H} + \sum_{i=j}^{n-1} S_{n-1,i;U} y^{i-j} \right)$$

for $0 \leq j \leq n - 1$. In particular, $S_{n,n;H} = 0$.

Using $S_{n,j;U}$ and $S_{n,j;H}$, we can provide a combinatorial interpretation of the coefficient $r_{n,j}$ in the Taylor remainder polynomial of the generating function $H(w)$ for HD-free Schröder paths.

Theorem 9. For every $n \geq 1$ and $0 \leq j \leq n + 1$, the coefficient $r_{n,j}$ in the Taylor remainder polynomial of the generating function $\mathbf{H}(w)$ for HD-free Schröder paths satisfies

$$r_{n,j} = S_{n,j-1} y^{j-1} - S_{n,j;H} y^j. \quad (6)$$

Proof. Recall the Taylor expansion of $\mathbf{H}(w)$ as stated in Theorem 6:

$$\mathbf{H}(w) = \sum_{k=0}^{n-1} \mathbf{h}_k w^k + w^n R_n(\mathbf{H}(w)), \quad R_n(u) = \sum_{j=0}^{n+1} r_{n,j} u^j.$$

Thus, the term $w^n R_n(\mathbf{H}(w))$ corresponds to the HD-free Schröder paths of semilength at least n , every such path must meet the line $u + v = 2n$.

Let P be an HD-free Schröder path of semilength at least n . We decompose it as $P = P_1 P_2$, where P_1 is the initial HD-free Schröder subpath that reaches the line $u + v = 2n$ for the first time; thus $P_1 \in \mathcal{S}(n, j)$ for some integer $0 \leq j \leq n$. The last step of P_1 is either an up step U or a horizontal step H , corresponding to $P_1 \in \mathcal{S}_U(n, j)$ and $P_1 \in \mathcal{S}_H(n, j)$, respectively, and P_2 consists of the remaining part of the path. The detailed decomposition of P is illustrated in Fig. 10: the left diagram corresponds to the case $P_1 \in \mathcal{S}_U(n, j)$, while the right diagram corresponds to $P_1 \in \mathcal{S}_H(n, j)$. In the latter case, due to the HD-free restriction, the first step of P_2 cannot be a down step D . This explains why, before the first down step D appears in P_2 , the corresponding generating function contribution is $\mathbf{H}(w) - 1$ rather than $\mathbf{H}(w)$.

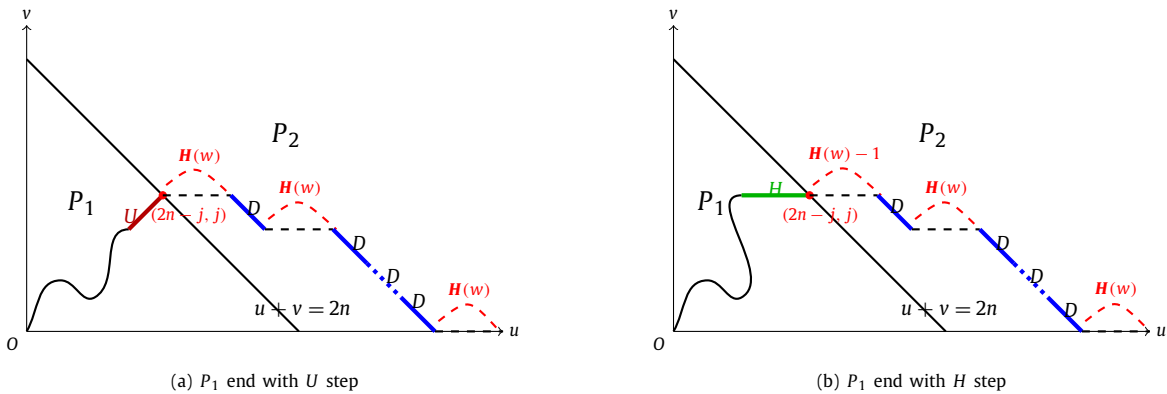


Fig. 10. Decompositions classified by the type of the last step taken before reaching the line $u + v = 2n$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Furthermore, to extract the coefficient $r_{n,j}$ from $w^n R_n(\mathbf{H}(w))$, we need to separate the contribution of a path P into the part that contributes to the factor w^n and the part that contributes to $R_n(\mathbf{H}(w))$. The factor w^n accounts for the semilengths of the steps of P_1 together with the j down steps in P_2 , which are highlighted in blue in Fig. 10. After removing these steps, the remaining portion of P consists of $j + 1$ separated components (all lying in P_2 , some of which may be empty). These components are obtained from P_2 by deleting the blue down steps indicated in Fig. 10. They contribute $\mathbf{H}(w)^{j+1}$ when $P_1 \in \mathcal{S}_U(n, j)$, and $(\mathbf{H}(w) - 1) \mathbf{H}(w)^j$ when $P_2 \in \mathcal{S}_H(n, j)$, for some integer $0 \leq j \leq n$.

In conclusion, we may decompose $R_n(\mathbf{H}(w))$ as

$$\begin{aligned} R_n(\mathbf{H}(w)) &= \sum_{j=0}^n S_{n,j;U} \mathbf{H}(w)^{j+1} y^j + \sum_{j=0}^n S_{n,j;H} (\mathbf{H}(w) - 1) \mathbf{H}(w)^j y^j \\ &= \sum_{j=0}^n (S_{n,j;U} + S_{n,j;H}) \mathbf{H}(w)^{j+1} y^j - \sum_{j=0}^n S_{n,j;H} \mathbf{H}(w)^j y^j \\ &= \sum_{j=0}^n S_{n,j} \mathbf{H}(w)^{j+1} y^j - \sum_{j=0}^n S_{n,j;H} \mathbf{H}(w)^j y^j. \end{aligned}$$

By extracting the coefficients of $\mathbf{H}(w)^j$ on both sides of the equation, we obtain (6). $\square$

For example, when $n = 2$ and $j = 0, 1, 2, 3$, we have

$$r_{2,0} = S_{2,-1} y^{-1} - S_{2,0;H} y^0 = -xyz - z^2,$$

$$\begin{aligned}r_{2,1} &= S_{2,0} y^0 - S_{2,1;H} y = z^2, \\r_{2,2} &= S_{2,1} y - S_{2,2;H} y^2 = 2xyz + x^2 y^2, \\r_{2,3} &= S_{2,2} y^2 - S_{2,3;H} y^3 = x^2 y^2,\end{aligned}$$

which matches the coefficients of $R_2(u)$ appearing after the proof of Theorem 6.

Recall that $r_{n,j}$ is a polynomial in the variables x, y , and z . Let $[x^a y^b z^c] r_{n,j}$ denote the coefficient of $x^a y^b z^c$ in $r_{n,j}$. We now proceed to give a combinatorial interpretation of the coefficient $[x^a y^b z^c] r_{n,j}$.

To represent consecutive horizontal steps H in a compressed form, we introduce $H^{(i)}$ to denote a segment consisting of i consecutive horizontal steps H , where i is a positive integer, such that the steps immediately before and after this segment (if they exist) are not H . Similarly, we define $D^{(i)}$ in the same manner. In particular, $H^{(0)}$ and $D^{(0)}$ mean that the corresponding segment of the path is absent.

The sign pattern of the coefficients of $r_{n,j}$ with respect to the variables x, y, z was previously observed from an algebraic perspective using generating functions. More precisely, the coefficients of $r_{n,0}$ in the variables x, y, z are nonpositive, whereas the coefficients of $r_{n,j}$ for $1 \leq j \leq n+1$ are nonnegative. Therefore, in what follows, we present the combinatorial interpretation of the coefficient $[x^a y^b z^c] r_{n,j}$ separately for the cases $j = 0$ and $j \geq 1$, which provides a concrete combinatorial explanation for the observed sign pattern of the coefficients of $r_{n,j}$.

Theorem 10. For every $n \geq 1$, the coefficient $[x^a y^b z^c] r_{n,0}$ is equal to the negative of the number of paths in $S_H(n, 0)$ having a up steps, b down steps, and c horizontal steps, where $a = b$ and $a + c = n$.

Proof. When $j = 0$, (6) reduces to $r_{n,0} = -S_{n,0;H}$. The statistics of the three types of steps follow directly from the definition of $S_{n,0;H}$. □

For example, when $n = 3$ and $j = 0$, we have $r_{3,0} = -2x^2 y^2 z - 2xyz^2 - z^3$. On the other hand, $S_H(3, 0) = \{UUDDH, UDUDH, UDHH, HUDH, HHH\}$. In $S_H(3, 0)$,

- There are exactly two paths with two up steps, two down steps and one horizontal step, namely $UUDDH$ and $UDUDH$, which corresponds to $a = b = 2$ and $c = 1$. Hence, the coefficient of $x^2 y^2 z$ is -2 .
- There are exactly two paths with one up step, one down step, and two horizontal steps, namely $UDHH$ and $HUDH$, corresponding to $a = b = 1$ and $c = 2$. Hence, the coefficient of xyz^2 is -2 .
- Finally, there is exactly one path with three horizontal steps, namely HHH , corresponding to $c = 3$. Hence, the coefficient of z^3 is -1 .

These agree with all the terms of $r_{3,0}$.

Theorem 11. For every $n \geq 1$ and $1 \leq j \leq n+1$, the coefficient $[x^a y^b z^c] r_{n,j}$ counts HD-free Schröder paths of semilength $n+1$ with $a+1$ up steps, $b+1$ down steps, and c horizontal steps, where $a = b$ and $a + c = n$. Moreover, such a path either ends with $U^{(i+1)} D^{(j)}$ or $UH^{(k)} UD^{(j)}$ for $2 \leq j \leq n+1$, and ends with $H^{(n)} UD$ when $j = 1$, where $i, k \geq 1$.

Proof. From the expression of $r_{n,j}$ given in (6), we obtain

$$\begin{aligned}r_{n,j} xy &= S_{n,j-1} xy^j - S_{n,j;H} xy^{j+1} \\&= S_{n,j-1;U} xy^j + S_{n,j-1;H} xy^j - S_{n,j;H} xy^{j+1}.\end{aligned}$$

Here, to properly handle the HD-free constraint, we consider the combinatorial interpretation of $r_{n,j} xy$ rather than that of $r_{n,j}$. Essentially, the difference is that the former corresponds to the addition of one extra up step U and one extra down step D .

Claim 11.1. All negative contributions from $-S_{n,j;H} xy^{j+1}$ are canceled by a part of positive contributions arising from $S_{n,j-1;H} xy^j$.

To see this combinatorially, let $P = P_1 H^{(i)} \in S_H(n, j)$ for some positive integer i , where $H^{(i)}$ denotes the maximal terminal segment consisting entirely of horizontal steps H . Then P_1 does not end with a horizontal step H . Define $P' = P_1 D H^{(i)}$. Then P' remains HD-free, and hence $P' \in S_H(n, j-1)$. Under this transformation, the number of down steps D in P' increases by one, while the numbers of all other types of steps remain unchanged. Thus, the proof of the claim is complete.

Claim 11.2. The term $S_{n,j-1} xy^j$ counts the paths obtained by taking a path in $S(n, j-1)$ and appending $UD^{(j)}$; these are precisely the HD-free Schröder paths of semilength $n+1$.

Let $P \in \mathcal{S}(n, j-1)$, namely, P meets the line $u+v=2n$ at the point $(2n-j+1, j-1)$. The factor xy^j corresponds to a segment consisting of an up step U followed by j consecutive down steps D . By concatenating these two parts, we obtain $PUD^{(j)}$, which finally reaches $(2n+2, 0)$. Hence, this construction yields an HD-free Schröder path of semilength $n+1$.

In the proof of Claim 1, part of the term $S_{n,j-1;H}xy^j$ has already been canceled by the negative contributions from $-S_{n,j;H}xy^{j+1}$. This cancellation corresponds precisely to those paths that end with a segment of the form $DH^{(i)}UD^{(j)}$ with some positive integer i . Consequently, the remaining contributions come from paths that either end with $U^{(i+1)}D^{(j)}$ or with $UH^{(k)}UD^{(j)}$ for $2 \leq j \leq n+1$, and from paths that end with $H^{(n)}UD$ when $j=1$, where $i, k \geq 1$.

Finally, the combinatorial interpretation of $[x^a y^b z^c]r_{n,j}$, together with the corresponding step-type statistics, follows. $\square$

For example, when $n=3$ and $j=1, 2$, we have

$$r_{3,1} = z^3 \quad \text{and} \quad r_{3,2} = 2x^3 y^3 + 3x^2 y^2 z + 3xyz^2.$$

For $j=1$, the unique path to be counted is $HHHUD$, which corresponds to $c=3$, and hence the coefficient of z^3 is 1. For $j=2$, the HD-free Schröder paths of semilength 4 satisfying the conditions described in Theorem 11 are

$$UDUDUDD, UUDDUDD, HUDUDD, UDHUDD, \\ UDUHDD, HHUDD, HUHDD, UHHDD.$$

Among them,

- There are exactly two paths with four up steps and four down steps, namely $UDUDUDD$ and $UUDDUDD$, which correspond to $a=b=3$. Hence, the coefficient of $x^3 y^3$ is 2.
- There are exactly three paths with three up steps, three down steps, and one horizontal step, namely $HUDUDD$, $UDHUDD$, and $UDUHDD$, corresponding to $a=b=2$ and $c=1$. Hence, the coefficient of $x^2 y^2 z$ is 3.
- There are exactly three paths with two up steps, two down steps, and two horizontal steps, namely $HHUDD$, $HUHDD$, and $UHHDD$, corresponding to $a=b=1$ and $c=2$. Hence, the coefficient of xyz^2 is 3.

These agree with the three terms of $r_{3,2}$.

If we ignore the terminal segment $UD^{(j)}$, an equivalent combinatorial interpretation of $[x^a y^b z^c]r_{n,j}$ can be formulated in terms of paths in $\mathcal{S}(n, j-1)$.

Theorem 12. For every $n \geq 1$ and $1 \leq j \leq n+1$, the coefficient $[x^a y^b z^c]r_{n,j}$ counts the number of paths in $\mathcal{S}(n, j-1)$ with a up steps, $b-j+1$ down steps, and c horizontal steps, subject to $a=b$ and $a+c=n$. Moreover, such a path ends either with U or with $UH^{(k)}$ for $2 \leq j \leq n+1$, while it ends with $H^{(n)}$ when $j=1$, where $k \geq 1$.

Consider again the example $r_{3,2}$, this time via the paths in $\mathcal{S}(3, 1)$ that satisfy the conditions of Theorem 12, namely

$$UDUDU, UUDDU, HUDU, UDHU, UDUH, HHU, HUH, UHH.$$

The correspondence between these paths and $[x^a y^b z^c]r_{3,2}$ follows in the same manner as in the argument preceding Theorem 12, we omit the details here.

6.2. Combinatorial interpretations of $f_{n,j}$

Similarly, the coefficient $f_{n,j}$ in the Taylor remainder polynomial of the generating function $\mathbf{S}(w)$ for skew Dyck paths can be interpreted in the same way after replacing z by xz . Consequently, we slightly modify the definitions of $S_{n,j}$ and $S_{n,j;H}$, and denote the resulting quantities by $\tilde{S}_{n,j}$ and $\tilde{S}_{n,j;H}$, respectively:

$$\tilde{S}_{n,j} := \sum_{P \in \mathcal{S}(n,j)} x^{u(P)+\ell(P)} y^{d(P)} z^{\ell(P)}$$

and

$$\tilde{S}_{n,j;H} := \sum_{P \in \mathcal{S}_H(n,j)} x^{u(P)+\ell(P)} y^{d(P)} z^{\ell(P)},$$

for integers $n \geq 1$ and $0 \leq j \leq n$.

Theorem 13. For every $n \geq 1$ and $0 \leq j \leq n+1$, the coefficient $f_{n,j}$ in the Taylor remainder polynomial of the generating function $\mathbf{S}(w)$ for skew Dyck paths satisfies

$$f_{n,j} = \tilde{S}_{n,j-1} y^{j-1} - \tilde{S}_{n,j;H} y^j.$$

The combinatorial interpretation of the coefficients of $f_{n,j}$ with respect to the variables x, y, z is given below.

Theorem 14. For every $n \geq 1$, the coefficient $[x^a y^b z^c] f_{n,0}$ is equal to the negative of the number of paths in $\mathcal{S}_H(n, 0)$ having $a - c$ up steps, b down steps, and c horizontal steps, where $a = b + c = n$.

Theorem 15. For every $n \geq 1$ and $1 \leq j \leq n + 1$, the coefficient $[x^a y^b z^c] f_{n,j}$ equals the number of HD-free Schröder paths of semilength $n + 1$ with $a - c + 1$ up steps, $b + 1$ down steps, and c horizontal steps, where $a = b + c = n$. Moreover, such a path either ends with $U^{(i+1)} D^{(j)}$ or with $UH^{(k)} UD^{(j)}$ for $2 \leq j \leq n + 1$, while it ends with $H^{(n)} UD$ when $j = 1$, where $i, k \geq 1$.

Theorem 16. For every $n \geq 1$ and $1 \leq j \leq n + 1$, the coefficient $[x^a y^b z^c] f_{n,j}$ equals the number of paths in $\mathcal{S}(n, j - 1)$ having $a - c$ up steps, $b - j + 1$ down steps, and c horizontal steps, subject to $a = b + c = n$. Moreover, such a path ends either with U or with $UH^{(k)}$ for $2 \leq j \leq n + 1$, while it ends with $H^{(n)}$ when $j = 1$, where $k \geq 1$.

6.3. Remarks on combinatorial interpretations via skew Dyck paths

We have already provided combinatorial interpretations of $f_{n,j}$ and $r_{n,j}$, as well as their coefficients in the variables x, y, z , in terms of HD-free Schröder paths. In principle, one may also obtain analogous combinatorial interpretations using skew Dyck paths. However, it is to be expected that such interpretations cannot be expressed in a clean and concise manner, mainly for the following reasons:

- Every HD-free Schröder path of semilength at least n necessarily intersects the line $u + v = 2n$, which allows us to define the sets $\mathcal{S}(n, j)$, a critical dividing line for the paths under consideration. However, due to the presence of left-down steps L , an n -skew Dyck path does not necessarily intersect the line $u + v = 2n$. Instead, its first possible intersection occurs at a shifted line of the form $u + v = 2n - \#L$, where $\#L$ denotes the number of left-down steps taken before the path first meets this line. Such a formulation is somewhat cumbersome and lacks the simplicity present in the HD-free Schröder path setting.
- For HD-free Schröder paths, after first meeting the line $u + v = 2n$, the path exhibits an intermittent descending behavior, where every descent is realized solely by a down step D . In contrast, in skew Dyck paths a descent can be achieved either by a down step D or a left-down step L . Consequently, when describing an intermittent descending behavior using skew Dyck paths, one must take into account the convolution of the step types D and L . Although explicit expressions can still be written down, the convolution structure prevents the resulting formulas from having a simple form.

For these reasons, given our preference for clear and transparent combinatorial interpretations, we do not pursue interpretations based on skew Dyck paths in detail here, although such interpretations do exist and can be obtained by analogous arguments.

7. Concluding remarks

The Taylor remainder polynomials of several well-known lattice path families, including Dyck paths, Motzkin paths, and Schröder paths, have attracted considerable attention in the literature. A common feature shared by these classes is that the coefficients of their Taylor remainder polynomials are nonnegative, which is consistent with the inherent nonnegativity of the underlying lattice-path statistics. This naturally leads to the expectation that the coefficients of the Taylor remainder polynomials of any generating function encoding counting statistics should always be nonnegative.

The starting point of this paper is to demonstrate that this expectation does not hold in general. We provide two concrete examples, skew Dyck paths and HD-free Schröder paths, whose Taylor remainder polynomials contain negative coefficients. Furthermore, we provide combinatorial interpretations for all coefficients of the Taylor remainder polynomials in both models, including the negative coefficients, thereby offering a combinatorial explanation for the observed sign patterns of the coefficients.

Along this direction, many related lines of study remain to be explored. In particular, one may expect that even more such examples exist. It is therefore natural to ask whether there is a general criterion that determines when negative coefficients appear, as well as to identify the structural factors governing the number of negative coefficients. Techniques for establishing log-convexity and related analytic properties of combinatorial sequences have been developed in [12], and may be useful in this context.

More broadly, existing research on Taylor remainder polynomials has focused almost exclusively on lattice-path models. The Taylor remainder polynomials arising from other types of combinatorial objects remain largely unexplored. Potential examples include Young tableaux, pattern-avoiding permutations, set partitions, and various classes of trees or graphs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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